

INTRODUCTION TO ECONOMICS

MGT404

Moral Hazard

Yale

Review and Today's Plan

Last class

Adverse selection is when one party is able to conceal their type, which increases costs (or, in the car case, lowers value) creating a pool that is *adversely selected*.

Strategies for overcoming:

- **BUYING INFORMATION:** Even when very costly, information can be worth it to acquire
- **INCREASE THE SURPLUS GAP:** The bigger the difference between cost and willingness to pay, the more “good” types can subsidize “bad”
- **SIGNALING:** Seller/customer can signal in a way that low-quality types will not replicate
- **SCREENING:** Firm sets up a menu of options that people self-select into
- **REPUTATION:** Repeated games (or legal system) can sustain higher equilibria

Today

Moral Hazard...

- Unobserved actions: principal vs agent
- Sales force compensation
- Efficiency wages
- Executive compensation

Introduction to “Moral Hazard”

Moral Hazard

In many situations, an “agent” making a decision will not bear the complete costs -- or not receive the full benefits -- of their efforts and/or actions.

- This is due to a mis-alignment between private incentives and global ones.
- This can result in an inefficient outcome.

For example, in the age of Covid...

- Some feel that getting vaccinated is privately costly
- The benefits of someone being vaccinated generally accrue to others (since you are less likely to transmit the disease to them)

From society’s point of view, everyone should be vaccinated, but some people choose not to as they see the private cost as greater than the private benefit.



This has a bit of the flavor of “externalities” from our lecture on taxes. The difference? There, we could contract on actions (if you pollute, you pay tax). Now, we don’t have perfect information over efforts or actions.

Moral Hazard

Principal-Agent Problems and Moral Hazard

Moral hazard problems arise when one person or entity, the **agent**, is tasked with taking actions on behalf of another party, the **principal**, and monitoring is imperfect. Typically, the agent has some cost of effort, while the principal benefits from their effort, and so the agent’s preferences might not match those of the principal.

Examples:	Principal	Agent	Tasks
Corporations	Shareholders	CEO and management	Maximize profits for shareholders.
Litigation	Client	Lawyer	Litigate for client’s best outcome (not just fees)
Politics	Voters	Politicians	Enact good laws, don’t be corrupt.
Influencers	Product sellers	Influencers	Give product information to maximize sales revenue

From Research:

[Return to Home Page](#) NATIONAL BUREAU of ECONOMIC RESEARCH

No Kin In The Game: Moral Hazard and War in the U.S. Congress

Eoin McGuirk, Nathaniel Hilger, Nicholas Miller

NBER Working Paper No. 23904

Issued in October 2017

NBER Program(s): DAE PE POL

Why do wars occur? We exploit a natural experiment to test the longstanding hypothesis that leaders declare war because they fail to internalize the associated costs. We test this moral hazard theory of conflict by compiling data on the family composition of 3,693 US legislators who served in the U.S. Congress during the four conscription-era wars of the 20th century: World War I, World War II, the Korean War, and the Vietnam War. We test for agency problems by comparing the voting behavior of congressmen with draft-age sons versus draft-age daughters. We estimate that having a draft-age son reduces legislator support for pro-conscription bills by 10-17%. Legislators with draft-age sons are more likely to be reelected subsequently, suggesting that support for conscription is punished by voters. Our results provide new evidence that agency problems contribute to political violence, and that elected officials can be influenced by changing private incentives.

Research finds that members of congress with draft-age sons vs draft-age daughters vote differently on conscription bills.



Moral Hazard in Health Insurance



“We’re just celebrating Joey got his health insurance back...”

Moral Hazard form Costly Effort

Principal-Agent Framework with Costly Effort

Many economic transactions (such as employment) can be thought of in the principal-agent framework where effort is involved:

- The **agent** is responsible for carrying out a task whose success depends on “effort”.
- The **principal** benefits from effort used on this task, but cannot observe “effort”

SOLUTION: The principal offers a wage contract that incentivizes the agent to exert effort at the lowest possible cost.

Examples:	Principal	Agent
Car Dealership	• Owner; wants sales staff to work hard.	• Sales staff: may rather be lazy.
Start-up	• Investors; want the project to succeed	• Entrepreneurs: have many projects.
Restaurant	• Owner; wants waiters to work hard.	• Waiters: want to do bare minimum.
Other examples?		

Electronics Retailers

The situation:

Traditional electronics retailers are struggling:

- Circuit City declared bankruptcy in 2009
- Target and Walmart expanded aggressively into electronics
- Consumers are using stores as “showrooms” for online purchases
- How does a firm like Best Buy survive?

One strategy: differentiate “up”



- Perks
- A showroom worth visiting
- Training, customer service

“...Abt pays employees a salary and bonuses, rather than hourly wages.”

Why?

Electronics Retailers Scramble to Adapt to Changing Market



An aquarium is one attraction at Abt Electronics in Glenview, Ill., where Tim Burke, left, helped Brandon Saposnik.

By STEPHANIE CLIFFORD
Published: June 18, 2012

Motivating a Sales Force

Suppose a **risk-neutral** (for now) salesperson at Abt can choose to “hide” or “help customers” while at work. A potential consumer comes by every hour, looking at TVs.

	Cost to Salesperson	Probability of Sale
Hide	0	n
Help	c	S

Exerting effort has a “cost”
to a salesperson.

Assume $S > n$, so that effort
increases the probability of a sale.

Key: If I see a customer did not purchase a TV, I don’t know if it was lack of effort or bad luck. Why is this hidden action? Effort is disguised by chance.

A salesperson’s “outside option” here is to go and work at Best Buy at \$ m per hour.

First-Best solution (perfect information)

If Abt could see the effort level, they could just offer a salary greater than $m + c$, and fire any employee that chose to “hide” instead of “help”.

The Problem

Unobserved Action

Unfortunately, it is not possible (or, extremely costly) to monitor whether each employee is “hiding” or “helping” every hour of their shift. Instead, all we see is whether or not a consumer decided to purchase a TV.

So, we can’t offer a wage based on “hide” vs. “help.” Instead, we could offer a wage based on customer purchases:

Consumer Decision:	Salesperson Salary
No purchase	Wage of w
Purchase	Wage of w plus bonus of b

Abt’s considerations

Abt needs to choose its wage and bonus considering two things: would an employee rather **work here** than at Best Buy (“willing to work”), and will an employee choose to **help** (“do the right thing”).

Constraints on Wages, Bonuses in Abt case

1. Wiling to Work

The expected wage/bonus must be such that I'm willing to work for Abt over my outside option.

$$\text{EV of "helping"} \geq \text{EV of working at Best Buy}$$
$$w + S \cdot b - c \geq m$$

Wage + prob of sale if work hard*bonus – cost of effort

Wage at next best job

2. Do the right thing

The wage and bonus must be such that it is worth it for an employee to “help”, in spite of the cost of effort.

$$\text{EV of "helping"} \geq \text{EV of "hiding"}$$
$$w + S \cdot b - c \geq w + n \cdot b$$

Wage + prob of sale if work hard*bonus – cost of effort

Wage + prob of sale if slack*bonus

Finding the best wage contract for Abt

Begin by paying just enough to make sure they are “willing to work”

I don't want to pay my employees more than I have to. So I choose my wage such that:

$$w^* + S \cdot b - c = m$$

i.e., I make it so they *just are* “willing to work”—their expected payoff from “helping” is just equal to their Best Buy wage.

This means the wage I pay will be:

$$w^* = m - S \cdot b + c$$

Now how do I structure that pay so they “do the right thing”?

I will choose my bonus to *just* incentivize my workforce, so I will choose b^* such that:

$$w^* + S \cdot b^* - c = w^* + n \cdot b^*$$

i.e., They're indifferent between helping and hiding—the bonus compensates their effort.

Subtract wage from both sides $\rightarrow S \cdot b^* - c = n \cdot b^* \rightarrow (S - n) \cdot b^* = c$

$$\rightarrow b^* = \frac{1}{S - n} \cdot c$$

Move b^* to one side so can solve for it.

Plugging in to the above formula, I can find the wage:

$$w^* = m + c - S \cdot \frac{c}{S - n} \rightarrow w^* = m - \frac{n}{S - n} \cdot c$$

What does it mean?

The optimal wage contract

Sales associates get w^* per hour, plus a bonus of b^* if they sell a TV:

$$w^* = m - \frac{n}{S-n} \cdot c \quad \rightarrow \quad b^* = \frac{1}{S-n} \cdot c$$

- As effort becomes more costly (as c increases), more of the pay has to be bonus pay relative to salary.
- If “help” is hard to distinguish from “hide” (i.e. $S - n$ is small), then b^* is big and w^* is small.

The big idea

- If a sales associate “helps”, they earn $w^* = m - \frac{n}{S-n} \cdot c$ with probability $1 - S$ and $w^* + b^* = m + \frac{1-n}{S-n} \cdot c$ with probability S .
- So, sales associates get their Best Buy wage m plus a gamble whose EV exactly compensates them for the cost of effort, c

- Unfortunately, this means even a hard-working sales associate is exposed to chance.
- If associates are risk averse, there may be an efficiency loss in compensating them this way.

Numerical Example

What does the optimal contract look like for reasonable parameters?

Best Buy Wage	$m = \$10$
Cost of Effort	$c = \$3$
S	$S = 75\%$
n	$n = 25\%$



For these parameters, the optimal contract is a bonus of

$$b^* = \frac{c}{S - n} = \frac{\$3}{50\%} = \$6$$

And an hourly wage of

$$\begin{aligned} w^* &= m - S \cdot b^* + c \\ &= \$10 - 75\% \cdot \$6 + \$3 \\ w^* &= \$8.50 \end{aligned}$$

Outcome:

If I “help”...

If I “hide”...

At ABT
(bonus)

- I get a wage of $w = \$8.50$ but my effort costs me $c = \$3$.
- With probability $S = 75\%$, I get a bonus of $b = \$6$.

$$\begin{aligned} EV &= \$8.50 - \$3 + (75\% \cdot \$6) \\ &= \$10 \end{aligned}$$

- I get a wage of $w = \$8.50$ and have no cost of effort.
- With probability $n = 25\%$, I get a bonus of $b = \$6$.

$$EV = \$8.50 + (25\% \cdot \$6) = \$10$$

At Best Buy
(no bonus)

- I get a wage of m regardless of whether I help or hide
- Since effort costs me c , I will hide
- I earn $m = \$10$ per hour, so $EV = \$10$.

Summary

Discussion of Outcome

This result implies that if Best Buy is offering \$10 an hour, we can get away with offering \$8.50 an hour, if we offer a \$6 commission on a sale.

- As n rises, we get lower w and higher b . Does that make sense?

What if workers are *at all* risk averse? How does that affect this?

- The base wage is a *sure thing*. It will be optimal to offer the highest w .

In order to get an employee to be willing to work at ABT and willing to “help” instead of hide, ABT must offer **at least** \$8.50 as a base wage with **at least** \$6 bonus from a sale.

How does the outcome change if effort is more costly? If Best Buy raises its wages?

$$w^* = m - \frac{n}{S - n} \cdot c \qquad b^* = \frac{1}{S - n} \cdot c$$

Is it worth incentivizing effort?

Does Abt's incentive scheme outperform Best Buy's?

The benefit is increased sales, but of course the cost is higher wages. Suppose Abt's margin on a TV is \$ X . Then, it prefers to incentivize when its hourly profit per salesman is greater than if it just paid \$ m per hour and let its workers "hide" (like Best Buy):

$$\begin{aligned}
 & \pi_{Abt} \geq \pi_{BB} \\
 & S \cdot (X - b^*) - w^* \geq nX - m \quad \leftarrow \begin{array}{l} \text{Higher probability} \\ \text{sell*net margin} \\ \text{accounting for} \\ \text{bonus - wage} \end{array} \quad \begin{array}{l} \text{Lower probability} \\ \text{sell*margin - wage} \end{array} \\
 & (S - n) \cdot X \geq S \cdot b^* - m + w^* \\
 & (S - n) \cdot X \geq S \cdot b^* - m + (m - S \cdot b^* + c) \quad \leftarrow \text{Sub in for the wage} \\
 & (S - n) \cdot X \geq c \\
 & X \geq \frac{c}{S - n}
 \end{aligned}$$

Want to solve for X , so move all X 's to one side

For our earlier values, this says that you're willing to do this as long as your profit per TV is at least \$6. (Shortcut: your increase in profit from effort should be more than the employee's cost of that effort).

What if workers are risk-averse?

Suppose workers have utility $u(x) = \ln(x)$.

1. Wiling to Work

$$\begin{aligned} \text{EU of "helping"} &\geq \text{EU of working at Best Buy} \\ S \cdot \ln(w + b - c) + (1 - S) \cdot \ln(w - c) &\geq \ln(m) \end{aligned}$$

The wage and bonus must be such that it is worth it for an employee to “help”, in spite of the cost of effort.

2. Do the right thing

$$\begin{aligned} \text{EV of "helping"} &\geq \text{EV of "hiding"} \\ S \cdot \ln(w + b - c) + (1 - S) \cdot \ln(w - c) \\ &\geq n \cdot \ln(w + b) + (1 - n) \cdot \ln(w) \end{aligned}$$

If workers are risk averse

- For risk-neutral workers, $w = \$8.50$ and $b = \$6$ can incentivize effort.
- Is this still true if workers are risk averse?

1. Wiling to Work

$$0.75 \cdot \ln(8.5 + 6 - 3) + (1 - 0.75) \cdot \ln(8.5 - 3) \geq \ln(10)$$

$$2.25 \not\geq 2.30$$

2. Do the right thing

$$0.75 \cdot \ln(8.5 + 6 - 3) + (1 - 0.75) \cdot \ln(8.5 - 3) \geq 0.25 \cdot \ln(8.5 + 6) + (1 - 0.25) \cdot \ln(8.5)$$

$$2.25 \not\geq 2.27$$

If workers are risk averse

- What if I raise only wage $w = \$9$ and leave bonus unchanged at $b = \$6$?

1. Wiling to Work

$$0.75 \cdot \ln(8.5 + 6 - 3) + (1 - 0.75) \cdot \ln(8.5 - 3) \geq \ln(10)$$

$$2.31 \geq 2.30$$

Does not work!

Wage increase affects utility under no effort too much

2. Do the right thing

$$0.75 \cdot \ln(8.5 + 6 - 3) + (1 - 0.75) \cdot \ln(8.5 - 3) \geq 0.25 \cdot \ln(8.5 + 6) + (1 - 0.25) \cdot \ln(8.5)$$

$$2.31 \geq 2.32$$

If workers are risk averse

- What if I only raise bonus $b = \$6.5$ and leave wage unchanged at $w = \$8.5$?

1. Wiling to Work

$$0.75 \cdot \ln(8.5 + 6 - 3) + (1 - 0.75) \cdot \ln(8.5 - 3) \geq \ln(10)$$

$$2.29 \not\geq 2.30$$

Does not work!
Not enough to compensate workers
for increased risk

2. Do the right thing

$$0.75 \cdot \ln(8.5 + 6 - 3) + (1 - 0.75) \cdot \ln(8.5 - 3) \geq 0.25 \cdot \ln(8.5 + 6) + (1 - 0.25) \cdot \ln(8.5)$$

$$2.29 \geq 2.28$$

If workers are risk averse

- What if I only raise bonus by a lot $b = \$7$ and leave wage unchanged at $w = \$8.5$?

1. Wiling to Work

$$0.75 \cdot \ln(8.5 + 6 - 3) + (1 - 0.75) \cdot \ln(8.5 - 3) \geq \ln(10)$$

$$2.32 \geq 2.30$$

This works!

But maybe we are leaving too much on the table.

2. Do the right thing

$$0.75 \cdot \ln(8.5 + 6 - 3) + (1 - 0.75) \cdot \ln(8.5 - 3) \geq 0.25 \cdot \ln(8.5 + 6) + (1 - 0.25) \cdot \ln(8.5)$$

$$2.32 \geq 2.29$$

If workers are risk averse

- Optimal solution: raise **both** wages and bonus.
 $w = \$8.725$ and $b = \$6.32$

1. Wiling to Work

$$0.75 \cdot \ln(8.725 + 6.32 - 3) + (1 - 0.75) \cdot \ln(8.725 - 3) \geq \ln(10)$$
$$2.30 \geq 2.30$$

2. Do the right thing

$$0.75 \cdot \ln(8.725 + 6.32 - 3) + (1 - 0.75) \cdot \ln(8.725 - 3) \geq 0.25 \cdot \ln(8.725 + 6.32) + (1 - 0.25) \cdot \ln(8.725)$$
$$2.30 \geq 2.30$$

When is this worth it?

Does worthiness of bonus pay depend on risk aversion?

$$\pi_{Abt} \geq \pi_{BB}$$

$$S \cdot (X - b^*) - w^* \geq nX - m$$

$$0.75 \cdot (X - 6.32) - 8.725 \geq 0.25X - 10$$

$$X^* = 6.93$$



With risk-neutral workers, we found that wage + bonus was worth it as long as your profit per TV is at least \$6. When workers are risk-averse, profits have to be higher. Intuition: we have to compensate workers for facing risk under a bonus compensation scheme.

Economics in Action: Safelite Auto Glass



“During 1994 and 1995, after the introduction of new management, the company gradually changed the compensation method for its workforce, moving them from hourly wages to piece-rate pay”

What changes might you expect?

From the paper:

1. A switch to piece-rate pay has a significant effect on average levels of output per worker. **This is in the range of a 44-percent gain.**
 - A. The gain can be split into two components. About half of the increase in productivity results from the **average worker producing more** because of incentive effects. Some of the increase results from an ability to **hire the most productive workers** and possibly from a reduction in quits among the highest output workers.
2. The firm shares the gains in productivity with its workforce. A given worker receives about a 10-percent increase in pay as a result of the switch to piece rates.

Edward P. Lazear, “Performance Pay and Productivity”, The American Economic Review, Dec 2000

Moral Hazard: Today's Plan

Today's plan:

Wage + Bonus

A salesforce may prefer to be lazy if effort is costly. If you cannot perfectly monitor, you can reward observed sales to incentivize effort.

Efficiency Wages

If employment is a repeated game, you can pay a premium to employees so that if they are caught not working, getting fired is very costly.

Executive compensation

Designing pay packages for executives is complex. How can you incentivize them to take decisions the shareholders would approve of? What if they are risk-averse?

Could Wal-Mart offer Sales Bonuses?

We solved for an approach to motivate employees using a bonus tied to sales. What are some reasons why this might not work for, say, Walmart?

Reasons

Monitoring

Notes

- It is difficult to track what sales resulted from a particular employee's actions.

Multi-tasking

- Many tasks that we want to encourage don't directly generate sales

Margins

- Profit margins are generally low. Might not make sense to tie particular sales to bonuses.



Gain from effort lower, harder to measure; monitoring is costly. When does just paying a higher base wage work?

Is Employment a One-Shot Game? Efficiency Wages

There are many hours in a work day:

If I “hide” once...

I might get away with it!

- At ABT, this saves me “\$3” in effort in an hour.

If I “hide” a lot ...

I could save \$3 every hour!

- I might also get fired...

Employment is a *repeated game*. You will study in game theory how firms can leverage *repeated interactions* to encourage cooperation.

Job Opening	Skills	Wage
McDonalds	• HS Education	• ω
Shake Shack	• HS Education + effort	• $W > \omega$ plus benefits

Is monitoring perfect? If I don't put in effort will I get fired right away? Eventually?
The risk of getting caught (and fired) means that you will exert effort.

Shake Shack vs. McDonald's



How is working at Shake Shack similar to McDonalds? How is it different?

Similar

- Fast food chains with similar job roles and skills
- Similar hours, hierarchy

Different

- Shake Shack pays a higher wage
- Shake Shack offers benefits (401k, monthly bonus, health coverage)
- Higher employee loyalty?

By paying higher wages, Shake Shack can be more selective in the employees it hires and retains.

Moral Hazard: Today's Plan

Today's plan:

Wage + Bonus

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Efficiency Wages

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Executive compensation

Designing pay packages for executives is complex. How can you incentivize them to take decisions the shareholders would approve of? What if they are risk-averse?

CEO Pay Package Features

Feature	Reasoning
Exit Package/Parachute	• This provides the CEO with “insurance”, a minimum outcome should performance not be as hoped. • This may be necessary to lure a quality executive from another, safer position.
Base Salary	• Another form of insurance: a minimum level of compensation
Bonuses	• Compensation tied directly to performance • Drawback: may not work if CEO is risk averse
Restricted Stock	• Cannot be sold until certain conditions are met • Similar to bonuses
Stock options	• Can be structured to increase risk-seeking behavior. • Downside: can lead to excessive risk-seeking behavior.

Conclusion: Moral Hazard More Broadly

Any situation where an agent doesn't take into account the "full" cost or benefit of decisions when making them.

- In this sense, similar to the "externality" concept we discussed earlier in course.

Setting

Politics

- Would politicians vote for same laws if personally affected vs not?

Insurance

- Would individuals make same decisions if insured vs not insured?

Corporations

- Would CEO make same decisions if firm were private vs public?



All of these
have potential
for moral
hazard.

Summary and Direction

Today

Moral Hazard:

- Sometimes, **agents** can take unobserved actions.
- **Principals** cannot write contracts based on these unobserved actions.
- Instead, they can write incentive contracts.

The two constraints to keep in mind are:

- Incentives: give your workers a reason to work hard
- Participation: Make sure they don't leave to go and work elsewhere.

These ideas go from front-line workers up to CEOs.